Animal Behavior Exam # 1

Evolution and Behavior Basics

60 points total

Welcome to the first exam!

The purpose of this exam is to test your ability to apply the evolutionary and behavioral concepts we have been practicing in the first three modules. I am interested in seeing your thinking, so make sure to document your reasoning as you answer each question. Many of the questions are open-ended, and there may be more than one correct answer. If I ask you to generate a hypothesis or design an experiment, I likely don't have a specific answer in mind. I am looking to see that your answer is **logically consistent** (does your conclusion follow from your premises?), **reasonable** (are your claims biologically plausible given what we know about how organisms work?) and **tractable** (if proposing a study, could it be conducted by a real research program with finite resources?).

This exam is open book, and you may refer to your notes, my slides, or the textbook chapters whenever you need. While you may use any resource which assists you with facts and definitions, I expect the thinking on this exam to be your own. Please do not work on these questions with other students in the course, and please do not use AI tools to generate answers.

Question 1 (15 points total)

For 60 years, the English Greenweavers living on Staten Island have been split into two isolated groups by the Freshest Kills landfill. Last summer, the landfill was transformed into a public park and Greenweavers from both the Eastern and Western lineages are now found throughout the area. The park contains manicured lawns (low food availability), areas of mixed scrub (moderate food availability), and more natural woodlands (high food availability). After catching and weighing adult birds found nesting across the park, your research team observes the pattern depicted in Figure 1.

1A. Describe a pattern in figure 1 which suggests a genetic component to body mass. (5 pts)

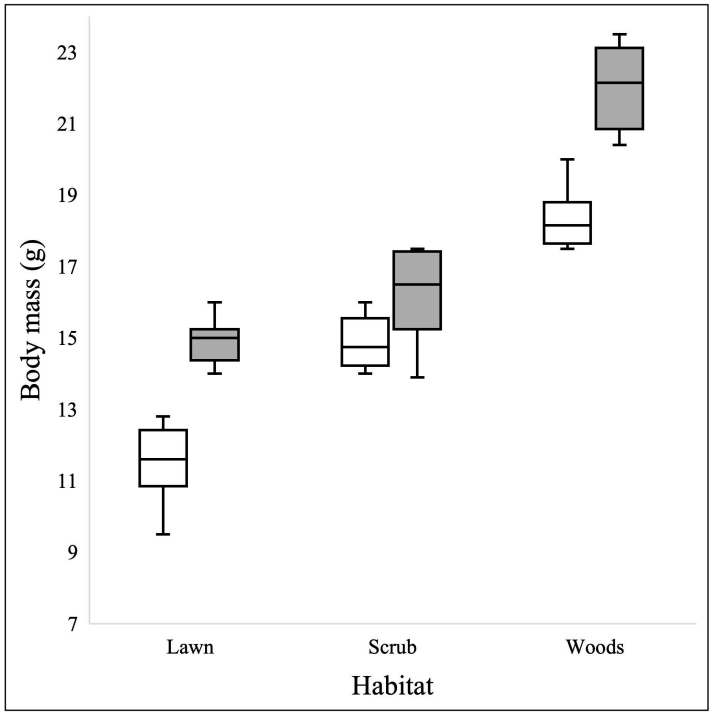
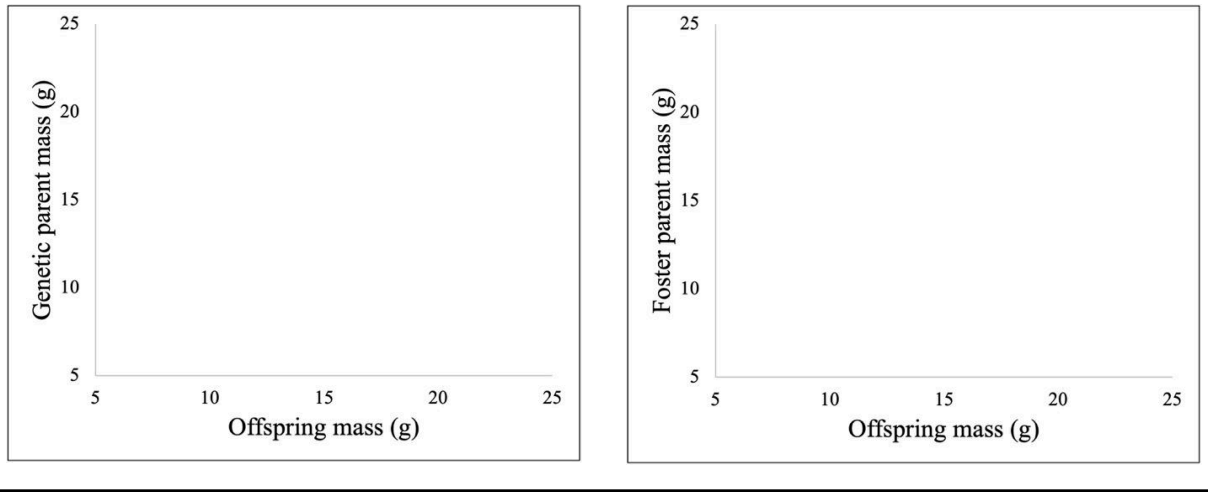


Figure 1: Adult greenweaver mass across three park sites for individuals from the western (white boxes) and eastern (grey boxes) lineages.

1B. Describe a pattern in figure 1 which suggests an environmental component to body mass. (5 pts)

While parental mass is generally correlated with offspring mass in this species, it is not known if this is due to underlying genetics. To test the heritability of body mass, you perform a cross-fostering experiment, swapping eggs between nests of Eastern lineage Greenweavers in the wooded site. Adult Greenweavers are not able to distinguish their eggs from those of other individuals, and thus raise all eggs in their nest as if they were their own. To your surprise, there is little evidence that body mass is heritable.

1C. Fill in the two panels below in a way that supports this conclusion. You can draw points or a trendline, provided your answer is clear. (2.5 pts)



1D. Offer an alternative non-heritable explanation for your proposed pattern in question 1A based on your answer to 1C. (2.5 pts)

Question 2 (15 points total)

You have been studying a population of New Mexico Reclining Jumping Spiders (*Habronattus laziboyus*). This species is found not only in lowland scrub desert, but also in mid-elevation montane forests and the bare rock of the mountain tops. These spiders are visual hunters, and jump significant horizontal distances to catch prey. They feed on a variety of ground-dwelling arthropods, most of which are present across all three habitats.

Your team collects females with egg sacs, counts the number of eggs in each sac, and brings the spiders to the lab to measure jump distance under standardized conditions[1]. Previous research has shown that the adult jump distance is consistent within an individual across all life stages. Using these data, you produce Figure 2, given below.

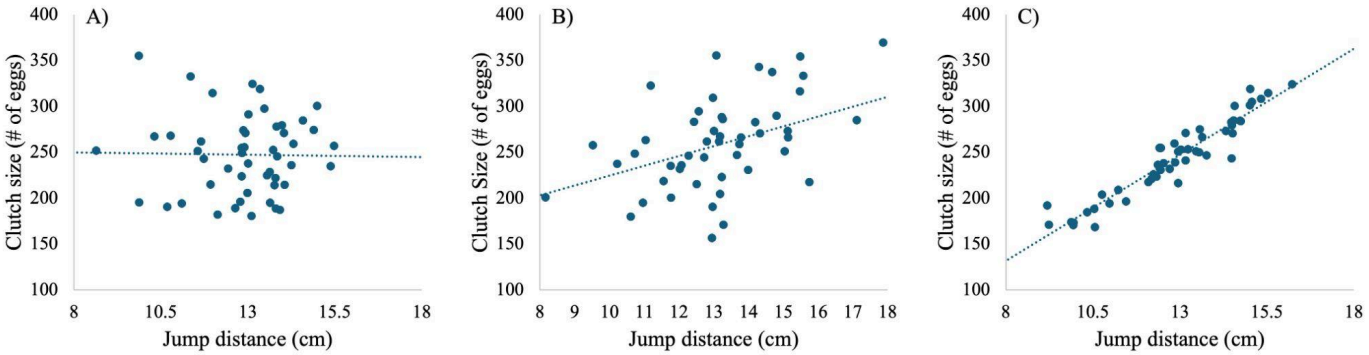


Figure 2: Jump distance and clutch size measured in female H. laziboyus at a) low-elevation desert, b) mid-elevation forest, and c) high elevation bare rock.

2A. Describe the relationship between elevation and selection on jump length. At which location is it strongest? In which direction is selection acting at that location? (5 pts)

[1] You put them on a graph-paper sheet and poke them in the butt with a paintbrush. No joke. This is how this sort of thing is done.

2B: Generate a reasonable hypothesis for why selection strength varies across these environments. (5 pts)

2C: Based on these patterns alone, what can you predict about how mean jump length will change over the next 50 generations across these sites? (5 pts)

Question 3 (15 points total)

You have been hard at work studying Eastern parcel bees (*Apis fedexeri*), a newly discovered species native to Northeastern North America[2]. Like their more famous honeybee cousins, parcel bees are smart, fun, and social! Mature colonies contain between 15,000 and 60,000 individuals, mostly workers. Unlike other honeybees, parcel bees visit only one flower species: the dry daisy. Dry daisies have unusually low pollen and nectar yields and are not preferred food sources for most other pollinators. They do, however, have unique ultraviolet markings on their petals which make them easy to identify at a distance (if you can see ultraviolet colors, which bees can).

Parcel bees are about half the mass of most other honeybees, and colonies grow faster and mature earlier in the season. Though, parcel bees are usually outcompeted by larger species when fighting over access to flower pollen. Parcel bees use a combination of sensory information (flower scent and, at closer range, color markings) and information sharing (workers perform a characteristic "post-classical interpretive dance-mime" to inform nestmates of the bearing and distance to food sources). They also create complex mental maps of the environment, navigating by use of both local landmarks and astronomical cues.

The average parcel bee worker lives for 4-6 months after maturing, and completes as many as 30 foraging trips per day, often journeying as far as 6 kilometers from the hive. This is considerably farther than other related bee species. Workers begin to actively forage within a day of maturing, and research has found that their foraging efficiency improves with age in several ways. Flights tend to become shorter, pollen yields higher, and mistakes (i.e., visitations to non-dry daisy flowers) less frequent.

Why are these bees smaller than usual? Why do they forage farther away than other species, and on less appealing flowers? What is up with their navigation, their communication, and their sensory systems? Why do bees get more efficient at foraging with age (or perhaps experience)?

Sometimes the problem is having too many questions to ask. And sometimes, the problem is that the question you're asking is not precise enough to be answered.

Q3A. On the next page, write one interesting[3], specific, and clear question about these bees for each of Tinbergen's four categories. (12.5 pts)

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[2] Not that it matters, but this is unusual. All other *Apis* species are native to Eurasia, likely arising in South Asia. Let me be clear - there is no hint or relevant information hidden in this footnote. You can disregard this factoid with complete confidence.

[3] Interestingness is yours to define — I'm curious what you find compelling about these bees, not whether you can guess what I find compelling. The only wrong answer here is a question so broad it can't be answered, or so narrow it doesn't matter.

Q3B: Which of these questions do you think would be most difficult to answer? What makes it difficult? (2.5 pts)

Question 4 (15 points total)

4A. Choose any of your four questions from part 3. Based on the natural history data provided, generate a reasonable testable hypothesis to answer your question. (2.5 pts)

4B. Design an experiment to test this hypothesis. Describe your protocols, treatments and measurements. Don't worry about picking a sample size or justifying equipment costs, etc. But do assume that you have the resources of a well-stocked behavioral ecology lab. (10 pts)

4C. Describe your predicted results if your hypothesis is supported and if it is not. (1 pt)

4D. Identify one methodological complication in your design which is genuinely difficult to solve, and discuss how you might deal with it in your discussion. (1.5 pts)